Laboratory 0:
An Introduction to the ECE 280 Labs!

Contents

Note on Prompts

Throughout this and future lab documents, there will be four different prompts to let you know specific work that needs to be done.

- **Pre-lab Deliverable (N):** Work to be documented in the pre-laboratory assignment.
- **Checkpoint (N):** Work to be shown to a TA during the laboratory.
- **Discussion (N):** Questions to be discussed with your lab partner and then answered in the post-laboratory assignment.
- **Deliverable (N):** Other items to be completed individually for the post-laboratory assignment.

1 Abstract

This semester, you will interface high-level software (MATLAB) with physical embedded hardware (Arduino Mega). To ensure a smooth "sampling pipeline" in future labs, your environment must be configured correctly. This lab guides you through the installation of MATLAB R2023b+, Arduino IDE 2.x, and the necessary hardware drivers to communicate with your Mega 2560.

2 Objectives

In this assignment, you will:

1. Locate the ECE 280 laboratory in Teer 116.
2. Verify MATLAB version compatibility and install required toolboxes.
3. Configure Arduino IDE 2.x with the correct Board Support Packages and Libraries.
4. Solve USB-to-Serial driver issues (the #1 cause of lab connectivity failures).

3 Pre-Laboratory Exercise

The ECE 280 labs meet in **Teer 116**.

- Enter Teer via the main entrance (Floor 2).
- Go down the open staircase to the Floor 1 Lobby.
- Pass the elevator and walk down the hallway past the EGR 201 lab.
- Take a right at the end of the corridor, past the Dean's portraits.

Pre-lab Deliverable (1/1): Take a selfie at the lab door and upload the image to the Gradescope assignment.

4 Laboratory Exercises

4.1 MATLAB Installation & Toolboxes

Required MATLAB version: R2020a minimum (R2023b+ strongly recommended).

Why this version floor matters:

- `exportgraphics()` was introduced in R2020a. Older versions fail figure-export deliverables.
- `serialport()`, `configureTerminator()`, and `readline()` were introduced in R2019b.
- Legacy `serial()` API is deprecated and not supported in our starter scripts.

Base MATLAB note: `fft`, `abs`, `plot`, `xlabel`, `ylabel`, `legend`, `zeros`, `ceil`, and `str2double` are included in base MATLAB.

How to verify your installation:

Table 1. MATLAB Toolboxes Required for ECE 280 Lab Sequence

Toolbox	Status	Why Needed / Typical Functions
Instrument Control Toolbox	**Required**	Serial communication with Arduino: `serialport`, `configureTerminator`, `readline`, `flush`.
Signal Processing Toolbox	**Required**	Filter/frequency analysis and report questions: `freqz`, `fir1`, `db2mag`.
DSP System Toolbox	**Required**	Streaming/spectrum workflows and advanced DSP tasks: `dsp.SpectrumAnalyzer`.
Control System Toolbox	**Required**	Transfer-function and response analysis: `bode`, `tf`, `step`.
Audio Toolbox	**Required**	Audio-oriented sampling/analysis extensions used in later labs.
Simulink	**Required**	Model-based signal and control workflows in upcoming labs.

```
ver
license('test','Instrument_Toolbox')
license('test','Signal_Toolbox')
license('test','DSP_System_Toolbox')
license('test','Control_Toolbox')
license('test','Audio_Toolbox')
license('test','Simulink')
```

Deliverable (1/3): Upload a screenshot of MATLAB Add-On Manager showing Instrument Control, Signal Processing, DSP System, Control System, Audio Toolbox, and Simulink installed.

4.2 Arduino IDE 2.x & Library Setup

Do **not** use legacy Arduino IDE 1.x. Install **Arduino IDE 2.3+** from https://arduino.cc/en/software.

Table 2. Board Support Package (Boards Manager)

Package	Version	Purpose
Arduino AVR Boards	1.8.6+	Required for Arduino Mega 2560 target support.

Install path:

- Boards package: Tools → Board → Boards Manager → search `Arduino AVR Boards`.
- TimerOne: Tools → Manage Libraries → search `TimerOne` → install.

Table 3. Library Manager Requirements

Library	Status	Purpose
TimerOne (Jesse Tane)	Recommended (install 1.1.0+)	Simplifies Timer1 configuration for high-frequency PWM workflows in Lab 1 and beyond.

4.3 USB Drivers & Port Permissions

Driver issues are the most common cause of failed serial communication.

Table 4. USB Driver by Board Variant

Board Variant	USB Chip	Driver Requirement
Official Arduino Mega 2560	ATmega16U2	Native CDC-ACM support on modern Windows/macOS/Linux (usually no manual install).
Most clone/off-brand Mega boards	CH340G	Manual CH340 driver install required.
Some clone boards	FTDI FT232	Usually auto-installed; FTDI vendor driver may improve stability.

Table 5. CH340 Driver Installation Notes

OS	Source	Notes
Windows 10/11	https://www.wch-ic.com	Run installer as Administrator and reboot.
macOS Ventura/Sonoma+	https://www.wch-ic.com	Approve kernel/security prompt in System Settings → Privacy & Security.
Linux (kernel 3.12+)	Built-in	Run `sudo usermod -aG dialout $USER`, then log out/in.

4.4 Finding the Correct Serial Port

4.5 Oscilloscope Screenshot Export (Optional)

If your scope supports PC transfer, you may use vendor software (e.g., Tektronix OpenChoice Desktop or equivalent) to export PNG/PDF screenshots for reports. A USB thumb drive save from the front panel is also acceptable.

4.6 Installation Checklist

Complete the following checklist before attending Lab 1.

Table 6. Port Discovery by Operating System

OS	**Where to Check**	**Typical Port Name**
Windows	Device Manager → Ports (COM & LPT)	COM3, COM4, ...
macOS	Terminal: `ls /dev/tty.*`	`/dev/tty.usbmodem*`, `/dev/tty.wchusbserial*`
Linux	Terminal: `ls /dev/ttyUSB*` `/dev/ttyACM*`	`/dev/ttyUSB0`, `/dev/ttyACM0`

- ☐ MATLAB R2020a or later installed (R2023b+ recommended).
- ☐ Instrument Control Toolbox installed and licensed.
- ☐ Signal Processing Toolbox installed and licensed.
- ☐ DSP System Toolbox installed and licensed.
- ☐ Control System Toolbox installed and licensed.
- ☐ Audio Toolbox installed and licensed.
- ☐ Simulink installed and licensed.
- ☐ Arduino IDE 2.x installed.
- ☐ Arduino AVR Boards package installed via Boards Manager.
- ☐ TimerOne library installed via Library Manager.
- ☐ Correct USB driver installed for your board (CH340 if clone).
- ☐ Port permissions configured (Linux dialout / macOS approval if needed).
- ☐ Port name identified and verified in MATLAB script settings.
- ☐ Arduino Serial Monitor tested at 115200 baud.
- ☐ Arduino IDE Serial Monitor **closed** before running MATLAB (single-app COM port access).

Checkpoint (1/1): Show your TA your completed MATLAB/Arduino installation and the functional Checklist.

Deliverable (2/3): Upload screenshots of: (1) MATLAB `ver` output, (2) toolbox license checks, and (3) Arduino IDE Boards/Library Manager pages showing AVR Boards and TimerOne.

4.7 MATLAB Tutorial

Follow the MATLAB Onramp tutorial: https://www.mathworks.com/learn/tutorials/MATLAB-onramp.html.

Deliverable (3/3): Upload your completed MATLAB Onramp Certificate PDF.

5 Laboratory Assignment

Submit the following to Gradescope:

- **Pre-lab Deliverable 1:** Lab Door Selfie.
- **Deliverable 1:** MATLAB Add-On Manager screenshot with all required toolboxes.
- **Deliverable 2:** MATLAB `ver` + toolbox license-check screenshots.
- **Deliverable 3:** Arduino IDE Board/Library Manager screenshots (AVR Boards + TimerOne).
- **Deliverable 4:** MATLAB Onramp Certificate PDF.

References and Inspiration

University Course Inspirations

This foundational lab prepares you for the full ECE 280 course by establishing the software and hardware infrastructure that all subsequent labs depend on:

- **MIT 6.003 (Signals, Systems, and Inference):** MIT's emphasis on rigorous mathematical analysis paired with practical implementation motivates the requirement for MATLAB—the industry-standard tool for signal processing education and research.
- **Georgia Tech ECE 2026 (Real-Time DSP):** Georgia Tech's real-time hardware-in-the-loop pedagogy drives the Arduino + MATLAB integration. The Arduino Mega 2560 is affordable, reliable, and capable of real-time DSP at kilohertz sample rates—ideal for teaching embedded signal processing on actual microcontroller hardware.
- **Rice University ELEC 241 (Real-Time DSP):** Rice's curriculum emphasizes **direct hardware implementation** on microcontrollers. This setup lab ensures you have a working embedded platform (Arduino Mega) before diving into signal acquisition, filtering, and real-time control in Labs 1–14.
- **Duke University ECE 280:** This course bridges classical Fourier analysis, digital signal processing, and modern communication systems with hands-on hardware validation. The setup ensures all students—regardless of home computer OS or prior Arduino experience—can successfully run labs in a consistent, reproducible environment.

Core Software References

- The MathWorks. *MATLAB Documentation* (https://www.mathworks.com/help/).
  - **Instrument Control Toolbox:** `serialport`, `readline`, `configureTerminator` for Arduino communication.
  - **Signal Processing Toolbox:** `fft`, `freqz`, `fir1`, `butter` for signal analysis and filter design.
  - **Control System Toolbox:** `tf`, `bode`, `step` for transfer function analysis.
  - **Simulink:** Block-diagram modeling for real-time system simulation and code generation.
- Arduino. *Arduino IDE Documentation* (https://docs.arduino.cc/).
  - **Arduino Mega 2560:** 54 digital I/O pins, 16 analog inputs, 4 hardware UARTs at up to 2 Mbps, 16 MHz crystal oscillator.
  - **Hardware Specifications:** Flash memory 256 KB, SRAM 8 KB, EEPROM 4 KB.
  - **IDE 2.x improvements:** Integrated debugging, library manager, Boards Manager for easy toolchain setup.

- WCH Semiconductor. *CH340 USB-to-Serial Adapter Datasheet and Driver* (https://www.wch-ic.com).
  - **Common issue:** Most Arduino clones use CH340G instead of the official ATmega16U2. Drivers must be manually installed on Windows/macOS; built-in on modern Linux kernels.
  - **Driver availability:** Cross-platform support (Windows, macOS Ventura/Sonoma, Linux kernel 3.12+).
- Analog Devices. *AD9833 Programmable Waveform Generator Datasheet.*
  - **Function generator reference:** Optional hardware for generating precision test signals in later labs.

Embedded Systems and Real-Time Fundamentals

- Vahid, F., & Givargis, T. (2010). *Embedded System Design: A Unified Hardware/Software Introduction* (2nd ed.). Wiley.
  - **Chapters 1–3:** Microcontroller architecture, I/O subsystems (GPIO, analog I/O), interrupt handling, and real-time scheduling.
  - **Arduino context:** Explains why the Mega's dual hardware UARTs, interrupt-driven ADC, and PWM outputs are critical for real-time DSP.
- Monk, S. (2013). *Arduino + MATLAB: Audio Waveform Generation, Signal Processing, and Visualization.*
  - **Practical guide:** Serial communication protocols, baud rate selection, buffering, and synchronization between Arduino and MATLAB.
- Atmel. *ATmega2560 Datasheet* (now Microchip Technology).
  - **Deep reference:** Hardware registers, timer configurations, ADC characteristics. Not required for Lab 0 but essential for advanced Labs 1–14.

Operating System Compatibility

- Microsoft. *Windows Device Manager and COM Port Configuration Guide* (https://docs.microsoft.com/).
  - **Windows 10/11:** Native CDC-ACM support for most Arduino boards; CH340 drivers improve clone board stability.
- Apple. *macOS Kernel Extension and Security Guidelines* (https://developer.apple.com/).
  - **Ventura/Sonoma:** User approval required in System Settings for third-party kernel extensions (CH340 drivers).
- Linux Foundation. *udev Rules and Device Permissions* (https://www.kernel.org/).

  - **dialout group:** Linux provides built-in USB-to-serial support; user permission configuration required.

Why This Setup Matters

This lab is foundational because **every subsequent lab depends on a working MATLAB + Arduino environment**. A broken serial connection, missing toolbox, or driver issue will block you from conducting the actual experiments. By investing time now in correct setup, you prevent weeks of frustration later.

The specific choices (MATLAB, Arduino Mega, 115200 baud, etc.) reflect decades of ECE education best practices: MATLAB is the standard in signal processing; Arduino Mega is affordable and reliable; and 115200 baud is the sweet spot between speed and USB reliability on breadboard circuits.

6 Updates and Changelog

Version history of Lab 0 setup requirements:

- **April 2026:** Updated for MATLAB R2023b, Arduino IDE 2.x, and CH340 driver requirements. Added References section with pedagogical foundations.
- January 2025: Updated lab location and enumerated add-ons.

For a detailed changelog of the entire ECE 280 Lab Redesign initiative, see CHANGELOG.md in the project root.